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YASHWANTH GOWDA S N

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EDUCATION

KS Institute of Technology (KSIT), Bengaluru **Expected May 2029**
B.E. in Computer & Communication Engineering CGPA: **8.52/10.0**

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, SQL, JavaScript
Backend & Distributed Systems: Spring Boot, Spring Cloud Gateway, Spring Security, FastAPI, REST APIs, WebSockets, Kafka
Frontend & Databases: React.js, HTML, CSS, MySQL, MongoDB, Redis, ChromaDB
AI/ML: TensorFlow, TensorFlow Lite Micro, INT8 Quantization, LangChain, RAG, Gemini API, OpenAI API, Ollama, Qwen3
Embedded Systems & IoT: ESP32, NXP FRDM-MCXN236, Arduino UNO Q, ESP-NOW, UART, SPI, TinyML
DevOps & Tools: Docker, Kubernetes, Git, Linux, Maven, Gradle, Postman
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems

PROJECTS

CarbonMesh v2 – AI-Powered Carbon Footprint Tracking & Reduction Platform GitHub
Arduino UNO Q, NXP FRDM-MCXN236, TinyML, INT8 Quantization, Qwen3-0.6B, Ollama, Python, HMAC-SHA256

- Developed an offline-first carbon intelligence platform using NXP FRDM-MCXN236 and an INT8-quantized TinyML model to classify daylight energy waste and after-hours loads, enabling local relay-based control.
- Integrated Qwen3-0.6B through Ollama for AI-driven carbon reduction insights, with deterministic carbon calculations and validation.
- Implemented HMAC-SHA256 authentication, monotonic sequence tracking, and a SHA-256 hash-chained ledger for secure, tamper-evident carbon records.

RESq Mesh – Edge AI & Autonomous Embedded System Telemetry Mesh GitHub
C/C++, Python, TensorFlow Lite Micro, INT8 Quantization, NXP FRDM-MCXN236, ESP32, ESP-NOW

- Developed an embedded network-health prediction system on NXP FRDM-MCXN236 (Cortex-M33) using an INT8-quantized TensorFlow Lite Micro model, achieving **10.3 ms inference latency** with a **13,572-byte tensor arena**.
- Implemented ESP32 peer-to-peer communication using ESP-NOW, offline state synchronization with causal sequence tracking, and HMAC-SHA256 integrity verification.

Cloud Native API Gateway
Java, Spring Boot, Spring Cloud Gateway, Redis, Kafka, Docker, Kubernetes, JWT, Resilience4j

- Developed a cloud-native API gateway with JWT-based authentication, Spring Cloud Gateway routing, and Redis-based rate limiting.
- Integrated Kafka for asynchronous communication and Resilience4j for fault tolerance; containerized the application using Docker for Kubernetes-ready deployment.

Persistent Memory AI Assistant
Python, FastAPI, LangChain, ChromaDB, RAG, Gemini API, OpenAI API, Docker

- Built a multi-session AI assistant using retrieval-augmented generation (RAG), ChromaDB vector search, and contextual memory retrieval to maintain conversational context across sessions.

ACHIEVEMENTS

- **1st Place among 102 teams** – HACKFEST 2026, 24-hour National-Level Hackathon, Sai Vidya Institute of Technology; awarded INR 20,000.
- **1st Place – Overall Winner** – MLH Hacktoberfest Hack Day Bengaluru 2026; awarded INR 10,000 and an Arduino UNO Q board.
- **1st Place** – ICARUS, CanSat Telemetry Analysis Competition, RV University.
- **2nd Place** – GitSetGo!, Open-Source Git/GitHub Relay Race, RV University.
- Received the **Suzlon Young Brilliance Award 2026**.
- **3rd Place** – EcoSpark 2026, for EcoSync AI.
- Solved **270+ LeetCode problems**; contest rating: **1587**.